Supplementary Material S4: Numerical Checks of Formal Results in Their Own Models

Recursive Training on Model-Generated Data: A Critical Integrative Review

of Failure Regimes, Adversarial Interactions and Governance

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S4. Numerical check of formal results in their own models

Purpose and scope. This supplement checks numerically, in the models in which they are proved, formal results summarised in Section 2.1 and Table 2 of the main text. It does not test whether these results transfer to transformer language models, which remains an open question (Section 7.5, priority 1). All runs use seed 20261029; the scripts and their raw output are available at the repository listed under Code Availability. Run time: 4.8 s on a single CPU core.

S4.1 Linear regression: replacement, accumulation and a fixed budget (Results 1–2). *Setting.* Ordinary least squares with $d = 10$, $N = 50$ samples per generation, noise variance $\sigma^2 = 1$, a fixed true parameter θ^* and isotropic Gaussian inputs drawn afresh at each generation[1]; at generation $g \geq 1$ labels are produced by the previous model of the same regime plus $\mathcal{N}(0, \sigma^2)$ noise; generations 0–20; $R = 300$ replicates. *Regimes. Replace*: fit on the new batch only. *Accumulate*: fit on the real data and all synthetic batches. *Accumulate-subsample*: fit on a uniform random subset of $2N$ points from the accumulated pool (a stand-in for a fixed training budget). *Outcome*: $E_g = \mathbb{E}\|\hat{\theta}_g - \theta^*\|^2$, normalised by $E_0 = \sigma^2 d/(N-d-1) = 0.256$.

Results (Table S4.1a, Fig. S4.1a). Under replacement the error follows $E_0(g+1)$ closely (maximum relative deviation 5.4% at $g = 1$, within the 95% CI). Under accumulation it stays below $E_0 \sum_{i=1}^{g+1} i^{-2}$ at every generation and below $E_0\pi^2/6 = 1.64\, E_0$ ($1.58 \pm 0.11\, E_0$ at $g = 20$). Under a fixed budget it exceeds $E_0\pi^2/6$ at $g = 4$ and reaches $2.05 \pm 0.12\, E_0$ at $g = 20$, still rising slowly: the bound of Result 2 requires every fit to use all retained data, not a fixed-size subsample.

S4.2 Gaussian mean estimation with a growing synthetic sample (Result 5). *Setting.* $\theta^* = 0$, $\sigma = 1$; initial real sample $n = 20$; at iteration $t \geq 2$ the estimate is the mean of $n_t = \lceil c_{t-1}\, n \rceil$ samples drawn from $\mathcal{N}(\hat{\theta}_{t-1}, 1)$; schedules $c_k = 1$, $c_k = k$ and $c_k = k^{1.5}$; $T = 200$; $R = 20{,}000$ replicates. In this model $n\,\mathbb{E}[(\hat{\theta}_T - \theta^*)^2] = 1 + \sum_{t=2}^{T} n/n_t$ exactly.

Results (Table S4.1b, Fig. S4.1b). With a constant schedule the error grows linearly; with a linear schedule it grows as $O(\log T)$ and is therefore unbounded; with $c_k = k^{1.5}$ it converges to a finite limit (observed 3.45 at $T = 200$, below $1 + \zeta(1.5) \approx 3.61$). The probability of improvement $P(T) = \Pr(|\hat{\theta}_T - \theta^*| < |\hat{\theta}_1 - \theta^*|)$ is below $\frac{1}{2}$ for every schedule and T: at $T = 2$ it is 0.355 ± 0.007, against the exact value $\frac{1}{2} - \pi^{-1} \arcsin(5^{-1/2}) \approx 0.352$. Under $c_k = k^{1.5}$ the error is bounded while $P(200) = 0.288 \pm 0.006$: a probability below $\frac{1}{2}$ is compatible with bounded error (Section 2.1, Result 5, "probability of improvement" remark).

What the supplement does not show. These simulations reproduce theorems in the models in which they are proved. They do not test whether Results 1, 2 or 5 hold for transformer language models, which is priority (1) of Section 7.5 of the main text.

[1]Whether Gerstgrasser et al. use fixed or fresh inputs is verified against their theorem statement; see main text, .

Table 1: Numerical check of Results 1, 2 and 5. Means shown with 95% confidence intervals where $R \geq 100$. All numbers from `sim/s4a_results.json`, seed 20261029.

(a) Linear regression, E_g/E_0; $d = 10$, $N = 50$, $R = 300$

g	Replace (vs $g+1$)	Accumulate (vs $\sum i^{-2}$)	Accsub ($2N$ budget)
1	2.11 ± 0.12 (2)	1.29 ± 0.09 (1.25)	1.28 ± 0.09
5	5.94 ± 0.30 (6)	1.48 ± 0.10 (1.49)	1.79 ± 0.10
20	20.70 ± 1.05 (21)	1.58 ± 0.11 (1.60)	2.05 ± 0.12

(b) Gaussian mean, $n\,E_T$ (sim / exact) and $P(T)$; $n = 20$, $R = 20{,}000$

Schedule	$T = 2$ (/ exact)	$T = 200$ (/ exact)	$P(200) \pm 95\%$
$c_k = 1$	1.99 / 2.00	200.0 / 200.0	0.044 ± 0.003
$c_k = k$	2.03 / 2.00	7.01 / 6.87	0.220 ± 0.006
$c_k = k^{1.5}$	1.97 / 2.00	3.45 / 3.47	0.288 ± 0.006

Figure S4.1. (a) Normalised excess risk in linear regression under replacement, accumulation and accumulate-subsample ($2N$ budget); dashed: $g + 1$ (Result 1); dotted: $\sum_{i \leq g+1} i^{-2}$ (Result 2); grey horizontal: $\pi^2/6$; bands: 95% CI over 300 replicates. (b) Normalised error of recursive Gaussian mean estimation under three synthetic-sample schedules (log–log); dashed: exact value; dotted: $1 + \zeta(1.5)$.

S4.3 Linear regression with a fixed input design (Results 1–2). *Setting.* As in S4.1 but one input matrix $X \in \mathbb{R}^{N \times d}$ per replicate is reused at every generation; $R = 300$. *Results* (Table S4.2a, Fig. S4.2c). Replacement follows $E_0(g + 1)$ closely (2.04 ± 0.12 at $g = 1$, 20.23 ± 1.07 at $g = 20$). Accumulation at $g = 20$ is $1.54 \pm 0.08\,E_0$; the theory value $\sum_{i=1}^{21} i^{-2} = 1.598$ lies within the 95% confidence interval $[1.46, 1.62]$, so the result is consistent with Theorem 2 of [1]. With fresh inputs (S4.1) accumulation remains below this value: the bound of (2) requires a fixed input design.

S4.4 Discrete distribution with a fixed sample size (Result 4). *Setting.* $K = 50$ outputs with $p_k \propto k^{-1.2}$; $N = 500$ samples per generation; $G = 100$ generations; $R = 300$. *Replace*: each generation is the empirical distribution of N samples drawn from the previous one. *Accumulate*: the original samples are kept and N new samples are added per generation. *Results* (Table S4.2b, Fig. S4.2a). Under replacement support falls from 46.0 to 6.8 outputs and the total-variation distance to p rises from 0.10 to 0.58. An output's survival time grows with its expected initial count $m_k = Np_k$ (correlation 0.97 across outputs with census rate below 20%) and exceeds m_k in 74.9% of output–replicate pairs, consistent with the lower bound of [3]; the ratio of survival time to m_k ranges from 7.0 to 4.1 (Table S4.2b), so the simulation does not show strict proportionality. Under accumulation the support cannot shrink, because the original samples are retained; the total-variation distance to p rises from 0.10 to 0.12 and then stabilises, consistent with bounded drift.

S4.5 Temperature-filtered sampling under accumulation (Result 6). *Setting.* Exact-tracking limit of [2] ($\delta = 0$): $K = 20$ outputs, 200 initial samples from $p_k \propto k^{-1.1}$, then 2,000 steps of $b = 10$ samples drawn from $q^{1/T_\mathrm{s}} / \sum_k q_k^{1/T_\mathrm{s}}$; $T_\mathrm{s} \in \{0.8, 1.0, 1.25\}$; $R = 200$. *Results* (Table S4.2c, Fig. S4.2b). With $T_\mathrm{s} = 1.0$ the distribution drifts and stabilises (total-variation distance between steps 1,000 and 2,000: 0.010; normalised

entropy 0.79 → 0.78), consistent with bounded drift (Theorem 1 of [2]). With $T_s = 1.25$ entropy rises towards 1 (0.79 → 0.95), i.e. towards the uniform distribution. With $T_s = 0.8$ mass concentrates (entropy 0.79 → 0.24; largest output share 0.31 → 0.82); no output reaches zero in finite time because counts only increase, which corresponds to approach towards the simplex boundary. These three regimes match those of Theorems 1–3 of [2].

Table 2: Numerical check of Results 1–2 (fixed design), 4 and 6. All numbers from `sim/s4b_results.json`, seed 20261029.

(a) Fixed-design regression, E_g/E_0 (95% CI); $d = 10$, $N = 50$, $R = 300$

g	Replace	$g+1$	Accumulate	$\sum_{i \le g+1} i^{-2}$
1	2.04 ± 0.12	2	1.23 ± 0.06	1.250
5	5.85 ± 0.31	6	1.45 ± 0.08	1.491
20	20.23 ± 1.07	21	1.54 ± 0.08	1.598

(b) Discrete distribution: support size / TV to p; $K = 50$, $N = 500$, $R = 300$

g	Replace	Accumulate ($T_s = 1$)
0	46.0 / 0.10	46.0 / 0.10
10	24.1 / 0.29	46.0 / 0.12
100	6.8 / 0.58	46.0 / 0.12

(c) Temperature-filtered accumulation after 20,000 additional samples; $K = 20$, $R = 200$

T_s	$H(q)/\log K$	TV to q_0	Largest share	TV steps 1k–2k
0.8	0.24	0.52	0.82	0.058
1.0	0.78	0.10	0.32	0.010
1.25	0.95	0.28	0.13	0.029

Figure S4.2. (a) Support size under replacement and accumulation ($K = 50$, $N = 500$; under accumulation support is fixed at its initial value by construction). Bands: 10–90% quantiles over replicates. (b) Normalised entropy $H(q)/\log K$ under temperature-filtered accumulation; dashed: uniform ($H = 1$); dotted: $H(p)$. Bands: 10–90% quantiles. (c) Fixed-design regression; dashed: $g+1$ (Result 1); dotted: $\sum_{i \le g+1} i^{-2}$ (Result 2); bands: 95% CI of the mean.

Figures

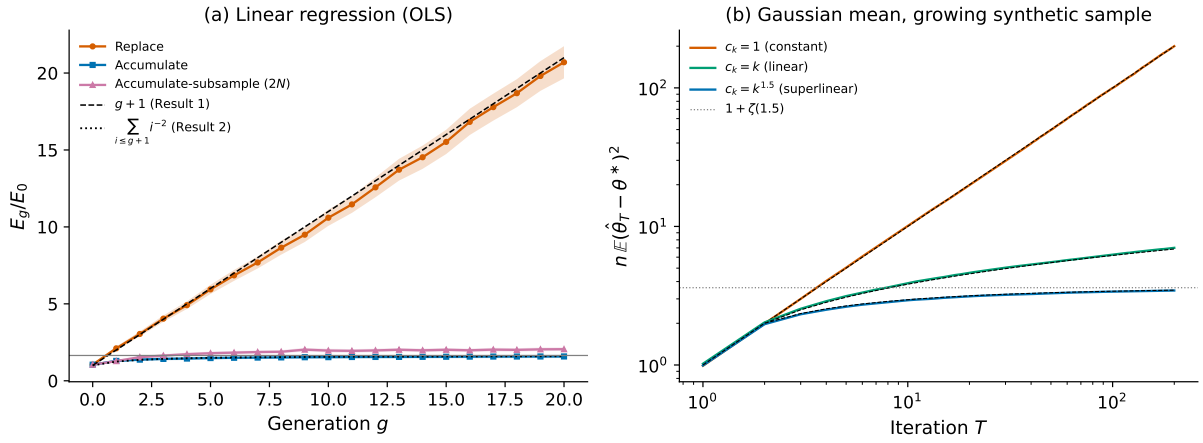


Figure 1: **Figure S4.1.** (a) Normalised excess risk in linear regression under replacement, accumulation and accumulate-subsample ($2N$ budget); dashed: $g+1$ (Result 1); dotted: $\sum_{i\le g+1} i^{-2}$ (Result 2); grey horizontal: $\pi^2/6$; bands: 95% CI over 300 replicates. (b) Normalised error of recursive Gaussian mean estimation under three synthetic-sample schedules (log–log); dashed: exact value; dotted: $1+\zeta(1.5)$. All numbers from `sim/s4a_results.json`, seed 20261029.

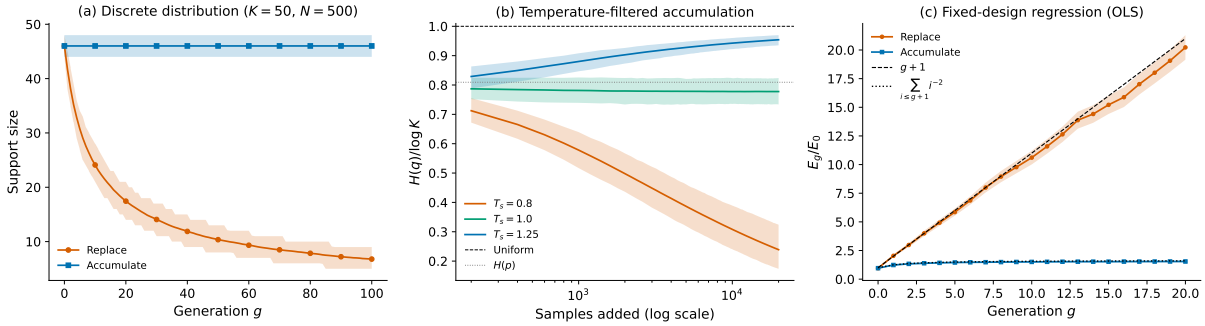


Figure 2: **Figure S4.2.** (a) Support size under replacement and accumulation ($K = 50$, $N = 500$; under accumulation support is fixed by construction). Bands: 10–90% quantiles. (b) Normalised entropy $H(q)/\log K$ under temperature-filtered accumulation; dashed: uniform ($H = 1$); dotted: $H(p)$. Bands: 10–90% quantiles. (c) Fixed-design regression; dashed: $g+1$ (Result 1); dotted: $\sum_{i\le g+1} i^{-2}$ (Result 2); bands: 95% CI. All numbers from `sim/s4b_results.json`, seed 20261029.

References

[1] Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Dhruv Pai, et al. Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. In *First Conference on Language Modeling*, 2024. URL `https://colmweb.org/2024/AcceptedPapers.html`.

[2] Matteo Marchi, Stefano Soatto, Pratik Chaudhari, and Paulo Tabuada. Heat death of generative models in closed-loop learning. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 1524–1530, Milan, Italy, 2024. IEEE. doi: 10.1109/CDC56724.2024.10886816.

[3] Ananda Theertha Suresh, Andrew Thangaraj, and Aditya Nanda Kishore Khandavally. Rate of model collapse in recursive training. In Yingzhen Li, Stephan Mandt, Shipra Agrawal, and Emtiyaz Khan, editors, *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 1396–1404. PMLR, 03–05 May 2025. URL `https://proceedings.mlr.press/v258/suresh25a.html`.